CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ SOLUTIONS

# Chapter 2: Polynomials

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

## 1 Zeroes and coefficients of a quadratic polynomial

**1.** $\alpha$ and $\beta$ are the zeroes of $p(x) = x^2 - 3x - 1$. Find $\frac{1}{\alpha} + \frac{1}{\beta}$. CBSE 2026, 2 marks

SOLUTION

$\alpha + \beta = 3$ and $\alpha\beta = -1$

$$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} = \frac{3}{-1}$$

**Answer:** $-3$

**2.** Assertion (A): The polynomial $p(y) = y^2 + 4y + 3$ has two zeroes. Reason (R): A quadratic polynomial has at most two zeroes. Choose: (a) both true and R explains A (b) both true but R does not explain A (c) A true, R false (d) A false, R true CBSE 2026, 1 mark

SOLUTION

$y^2 + 4y + 3 = (y + 1)(y + 3)$, so the zeroes are $-1$ and $-3$. A is true.

R is also true. But "at most two" only says there cannot be more than two zeroes. It does not show that this polynomial has exactly two.

**Answer: (b)**

**3.** The zeroes $\alpha$ and $\beta$ of the polynomial $3x^2 + 6x + k$ satisfy $\alpha + \beta + \alpha\beta = -\frac{2}{3}$. Find $k$. CBSE 2025, 1 mark

SOLUTION

$$\alpha + \beta = -\frac{6}{3} = -2 \text{ and } \alpha\beta = \frac{k}{3}$$

$-2 + \frac{k}{3} = -\frac{2}{3}$, so $\frac{k}{3} = \frac{4}{3}$

**Answer:** $k = 4$

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**4.** Find the zeroes of the polynomial $x^2 + \frac{4}{3}x - \frac{4}{3}$. CBSE 2025, 2 marks

SOLUTION

$x^2 + \frac{4}{3}x - \frac{4}{3} = \frac{1}{3}(3x^2 + 4x - 4)$

$3x^2 + 4x - 4 = 3x^2 + 6x - 2x - 4 = 3x(x+2) - 2(x+2) = (3x-2)(x+2)$

The polynomial is zero when $3x - 2 = 0$ or $x + 2 = 0$.

Check: sum $= \frac{2}{3} - 2 = -\frac{4}{3}$ and product $= \frac{2}{3} \times (-2) = -\frac{4}{3}$, which match the coefficients.

**Answer:** $\frac{2}{3}$ **and** $-2$

**5.** The zeroes of $p(x) = 2x^2 - k\sqrt{2}\,x + 1$ add up to $\sqrt{2}$. Find $k$. (a) $\sqrt{2}$ (b) 2 (c) $2\sqrt{2}$ (d) $\frac{1}{2}$

CBSE 2024, 1 mark

SOLUTION

Sum of zeroes $= -\frac{b}{a} = \frac{k\sqrt{2}}{2}$

$\frac{k\sqrt{2}}{2} = \sqrt{2}$ gives $k = 2$

**Answer: (b)**

**6.** Each zero of $x^2 + px + q$ is double a zero of $4x^2 - 5x - 6$. Find $p$. (a) $-\frac{5}{2}$ (b) $\frac{5}{2}$ (c) $-5$ (d) 10

CBSE 2024, 1 mark

SOLUTION

Let the zeroes of $4x^2 - 5x - 6$ be $\alpha$ and $\beta$. Then $\alpha + \beta = \frac{5}{4}$.

The zeroes of $x^2 + px + q$ are $2\alpha$ and $2\beta$, with sum $2(\alpha + \beta) = \frac{5}{2}$.

Sum of zeroes $= -p$, so $p = -\frac{5}{2}$

**Answer: (a)**

**7.** $\alpha$ and $\beta$ are the zeroes of $p(x) = x^2 + x - 1$. Find $\frac{1}{\alpha} + \frac{1}{\beta}$. (a) 1 (b) 2 (c) $-1$ (d) $-\frac{1}{2}$ CBSE 2023, 1 mark

SOLUTION

$\alpha + \beta = -1$ and $\alpha\beta = -1$

$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} = \frac{-1}{-1} = 1$

**Answer: (a)**

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**8.** Which of these is a quadratic polynomial with zeroes $-\frac{2}{3}$ and $\frac{2}{3}$? (a) $4x^2 - 9$ (b) $\frac{4}{9}(9x^2 + 4)$ (c) $x^2 + \frac{9}{4}$ (d) $5(9x^2 - 4)$ CBSE 2023, 1 mark

SOLUTION

Sum of zeroes $= 0$ and product $= -\frac{4}{9}$, so the polynomial is $k\left(x^2 - \frac{4}{9}\right) = \frac{k}{9}(9x^2 - 4)$.

Taking $k = 45$ gives $5(9x^2 - 4)$.

**Answer: (d)**

**9.** The sum of the zeroes of $x^2 - (k+6)x + 2(2k-1)$ is half of their product. Find $k$. CBSE 2019, 3 marks

SOLUTION

Sum of zeroes $= k + 6$ and product of zeroes $= 2(2k - 1)$

$k + 6 = \frac{1}{2} \times 2(2k - 1) = 2k - 1$

**Answer:** $k = 7$

## 2 Zeroes from the graph

**10.** The graph of $y = f(x)$ is a curve that stays above the $x$-axis everywhere and never touches it. How many zeroes does $f(x)$ have? (a) 0 (b) 1 (c) 2 (d) 4 CBSE 2026, 1 mark

SOLUTION

The zeroes are the $x$-coordinates of the points where the graph meets the $x$-axis.

This graph never meets the $x$-axis.

**Answer: (a)**

**11.** The graphs of two quadratic polynomials are drawn on the same axes. One parabola opens upwards and the other opens downwards, and both cut the $x$-axis at the same two points. How many distinct zeroes do the two polynomials have between them? CBSE 2025, 1 mark

SOLUTION

The zeroes of a polynomial are the $x$-coordinates of the points where its graph meets the $x$-axis.

Both graphs meet the $x$-axis at the same two points, so they share the same two zeroes.

**Answer: 2**

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**12.** Assertion (A): a polynomial whose graph touches the $x$-axis at exactly one point cannot be quadratic. Reason (R): a polynomial of degree $n$, with $n > 1$, has at most $n$ zeroes. Choose: (a) A and R are both true and R explains A. (b) A and R are both true but R does not explain A. (c) A is true and R is false. (d) A is false and R is true. CBSE 2024, 1 mark

SOLUTION

A is false: the graph of $(x-1)^2 = x^2 - 2x + 1$ touches the $x$-axis only at $x = 1$, and it is a quadratic polynomial.

R is true: a polynomial of degree $n$ has at most $n$ zeroes.

**Answer: (d)**

**13.** The graph of $y = p(x)$ is a parabola that opens downwards and touches the $x$-axis at exactly one point, without crossing it. How many zeroes does $p(x)$ have? (a) 3 (b) 1 (c) 2 (d) 0 CBSE 2023, 1 mark

SOLUTION

The graph meets the $x$-axis at only one point, so there is only one value of $x$ for which $p(x) = 0$.

**Answer: (b)**

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